

Problem

A 4-mF capacitor has the terminal voltage

$$v = \begin{cases} 50 \text{ V}, & t \leq 0 \\ Ae^{-100t} + Be^{-600t} \text{ V}, & t \geq 0 \end{cases}$$

If the capacitor has an initial current of 2 A, find:

(a) the constants A and B ,

(b) the energy stored in the capacitor at $t = 0$,

(c) the capacitor current for $t > 0$.

Step-by-step solution

Step 1 of 5

Given voltage equation

$$v = \begin{cases} 50 \text{ V}, & t \leq 0 \\ Ae^{-100t} + Be^{-600t} \text{ V} & t \geq 0 \end{cases}$$

Step 2 of 5

(a)

Now current flowing through the capacitor

$$i = C \frac{dv}{dt}$$

$$i = C(-100Ae^{-100t} - 600Be^{-600t})$$

Given $i(0) = 2 \text{ A}$ and $v(0) = 50 \text{ V}$

$$2 = 4 \times 10^{-3}(-100A - 600B)$$

$$100A + 600B = -500$$

$$A + 6B = -5 \dots\dots\dots (1)$$

Step 3 of 5

From voltage equation

$$A + B = 50 \dots\dots\dots (2)$$

From equations (1) and (2)

$$50 - B + 6B = -5$$

$$5B = -55$$

$$\boxed{B = -11}$$

From equation (2)

$$A = 50 + 11$$

$$\boxed{A = 61}$$

Step 4 of 5

(b)

$$\text{Energy} = \frac{1}{2} C [V(0)]^2$$

$$\text{Energy} = \frac{1}{2} \times 4 \times 10^{-3} [50]^2$$

$$\boxed{\text{Energy} = 5\text{J}}$$

Step 5 of 5

(c)

We know the current equation as

$$i = C(-100Ae^{-100t} - 600Be^{-600t})$$

$$i = 4 \times 10^{-3} (-100 \times 61e^{-100t} - 600 \times -11e^{-600t})$$

$$i = -24.4e^{-100t} + 26.4e^{-600t}$$

$$\boxed{i = 26.4e^{-600t} - 24.4e^{-100t}}$$